Curriculum Vitae

Taiyang Xu
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Citizenship: Chinese
Marital status: married

Employment

Aug 2026 – now	**Université catholique de Louvain**, Louvain-la-Neuve, Belgium Institut de Recherche en Mathématique et Physique Postdoctoral Researcher (Mentor: Prof. Christophe Charlier) Member of group *Random matrices, tilings, and water waves*
Jul 2024 – Jun 2026	**Fudan University**, Shanghai, China Department of Mathematics Postdoctoral Fellow (Mentor: Prof. Lun Zhang)

Education

Sep 2019 – Jun 2024	**Ph.D. in Mathematics** **Fudan University**, Shanghai, China Supervisor: Prof. Engui Fan Thesis title: "*On the long-time asymptotics of the local and nonlocal mKdV equation under the nonzero background*"
Sep 2015 – Jun 2019	**B.Sc. in Mathematics**, Distinguished Honor **China University of Mining and Technology**, Xuzhou, China Thesis title: "*Inverse scattering theory and integrability on several kinds of nonlinear evolution equations*"

Research Interests

Integrable ODEs/PDEs, Random matrix theory, Determinantal point processes, Orthogonal polynomials, Asymptotic analysis, Riemann-Hilbert problems, Special functions ...

Publications and Preprints

Listed in reverse chronological order of first public availability.

10. **Moment generating function of the tacnode process**
    arXiv:2606.17771

9. **Critical asymptotics of the Camassa-Holm equation in the small dispersion limit at the point of gradient catastrophe** (with Yiling Yang and Lun Zhang)
    available upon request

8. **Confluent hypergeometric kernel determinant on multiple large intervals** (with Lun Zhang and Zhengyang Zhao)
    arXiv:2508.10463

7. **Transient asymptotics of the modified Camassa-Holm equation** (with Yiling Yang and Lun Zhang)
    Journal of the London Mathematical Society, 110 (2024), e12967. / arXiv:2308.06950

6. **Painlevé transcendents in the defocusing mKdV equation with non-zero boundary conditions** (with Engui Fan and Zhaoyu Wang)
    Communications in Mathematical Physics, 406 (2025), 181. / arXiv:2306.07073

5. **Large-time asymptotics to the focusing nonlocal modified Kortweg-de Vries equation with step-like boundary conditions** (with Engui Fan)
    Studies in Applied Mathematics, 150 (2023), 1217–1273. / arXiv:2208.01268

4. **On the large-time asymptotics of the defocusing mKdV equation with step-like initial data** (with Yidan Zhang)
    Mathematical Physics, Analysis and Geometry, 29 (2026), 38. / arXiv:2204.01299

3. **On the Cauchy problem of defocusing mKdV equation with finite density initial data: long-time asymptotics in soliton-less regions** (with Engui Fan and Zechuan Zhang)
    Journal of Differential Equations, 372 (2023), 55–122. / arXiv:2108.06284

2. **Soliton resolution and asymptotic stability of N-soliton solutions for the defocusing mKdV equation with a non-vanishing background** (with Engui Fan and Zechuan Zhang)
    Physica D: Nonlinear Phenomena, 472 (2025), 134526. / arXiv:2108.03650

1. **Riemann-Hilbert approach for multisoliton solutions of generalized coupled fourth-order nonlinear Schrödinger equations** (with Weiqi Peng and Shoufu Tian)
    Mathematical Methods in the Applied Sciences, 43 (2020), 865–880.

Fellowships and Grants

Jul 2024 – Jun 2026	Shanghai Post-doctoral Excellence Program (Grant No. 2024100) Project: Riemann-Hilbert method for several asymptotic problems related to universality from integrable systems and random matrix theory Role: Principal Investigator
Jul 2024 – Jun 2026	China Postdoctoral Science Foundation (Grant No. 2024M760480) Project: Semiclassical asymptotics and universality for nonlinear integrable shallow water wave systems Role: Principal Investigator

Awards

Sep 2019 – Jun 2024 Fudan (Doctorate)

- Shanghai Outstanding Graduate (Graduation with Honors), 2024.
- Huatai Securities Science and Technology Scholarship, 2023.
- Pacific Insurance Company Scholarship, 2022.
- Outstanding Doctoral Candidate Scholarship, Fudan University, 2021.
- Fudan University Doctoral Scholarship, 2019–2023.

Sep 2015 – Jun 2019 CUMT (Undergraduate)

- Outstanding Undergraduates in China University of Mining and Technology, 2019.

Teaching

Fudan University (During Sep 2019 – Jul 2026)

- Fall 2025: Recitation & Tutorial Instructor of Mathematical Analysis III (MATH130001.04).
- Spring 2024: TA of Methods of Asymptotic Analysis (MATH630117).
- Fall 2021: TA of Calculus A (MATH120021.02).
- Spring 2020: TA of Calculus B, Online (MATH120004.01) .
- Fall 2019: TA of Calculus B (MATH120003.01).

Presentations

Research talks

- 2nd meeting of the FNRS contact group devoted to probability theory, Université libre de Bruxelles, Brussel, Belgium, 15–16 Oct, 2026.
  Large gap asymptotics for the confluent hypergeometric kernel determinant (invited)
- Seminar @Jiangsu Normal University, Xuzhou, China, 23 Jun, 2026.
  On the Fredholm determinants (invited)
- 3rd Workshop on Integrable Systems and Random Matrix Theory 2026, Dongguan, China, 4–17 Jan, 2026.
  Dubrovin's conjecture for the Camassa-Holm equation (invited)
- Bielefeld-Melbourne-Seoul Online Random Matrix Seminar, 15th Oct, 2025.
  Confluent hypergeometric kernel determinant on multiple large intervals (invited)
- 5th ZiF Summer School "Randomness in Physics and Mathematics: From Thermalisation in Quantum Systems to Random Matrices", Universität Bielefeld, Bielefeld, Germany, 25 Aug–6 Sep, 2025.
  Confluent hypergeometric kernel determinant on multiple large intervals (contributed)
- Seminar @UCLouvain, Louvain-la-Neuve, Belgium, 21 Aug, 2025.
  An introduction to Dubrovin's universality conjecture for integrable PDEs, and some progress on the Camassa-Holm equation (invited)
- The 2nd Workshop on Integrable Systems and Random Matrix Theory, Dongguan, China, 5–17 Jan, 2025.
  Transient asymptotics of the modified Camassa-Holm equation (invited)
- The 15th Hemudu Forum on Integrable Systems, Ningbo, China, 24–26 Nov, 2023.
  Integrable PDEs with nonzero boundary conditions: large-time asymptotics (contributed)

Outreach talks

- Shanghai Institute of Technical Physics, Shanghai, China, 29th Apr, 2025.
  Some asymptotic problems in mathematical physics (invited)

Others

12 **presentations** (including reading reports) given at Fudan Integrable Systems and Random Matrix Theory Seminar.

Academic Visits

22/06/2026 – 25/06/2026 Jiangsu Normal University, China. (Host: Dong Yao)

18/08/2025 – 24/08/2025 UCLouvain, Belgium. (Host: Christophe Charlier)

31/03/2025 – 11/04/2025 Chongqing University, China. (Host: Yiling Yang)

Community Services

Co-organized activities

- (with Lun Zhang) Weekly RMT seminar @Fudan, Sep 2022 – Jun 2026.
- (with Junwen Liu and Lun Zhang) Mini-workshop on Random Matrices, Fudan University, Shanghai, China, 4 June, 2026.
- (with Engui Fan, Zhaoyu Wang, Yiling Yang and Lun Zhang) Workshop on Integrable Systems, Fudan University, Shanghai, China, 28 Nov, 2025.
- (with Lun Zhang) Mini-workshop on Asymptotic Analysis, Fudan University, Shanghai, China, 5–9 Jun, 2025.

Referee services provided for the following journals

- Analysis and Applications
- Journal of Nonlinear Mathematical Physics
- Qualitative Theory of Dynamical Systems
- Studies in Applied Mathematics

Other services

- zbMATH Open Reviewer

Languages

- Chinese (native), The Qianwei dialect of Sichuanese (native)
- English

Computer Skills

LaTeX, Mathematica, Matlab, HTML, Python, Javascript

November 2026